Skills Summary

Function Notation, Domain, and Range

Use this reference while completing your worksheet or when studying.

One function, four faces. A rule, a table, a graph and a story can all describe the same function. Whichever you are handed, the notation is the same: $f(\text{input}) = \text{output}$.

Rule: $f(x) = 4x - 7$ • **Table:** an input row and an output row • **Graph:** the point (a, b) means $f(a) = b$

Domain = the inputs you are allowed to use. **Range** = the outputs they produce.

1. Evaluating a rule at an input

$f(3)$ is an instruction, not a product: write the rule again with the input in place of every x, then simplify. Put a negative input in parentheses first — $(-2)^2 = 4$, while $-2^2 = -4$.

Example: $f(x) = 5x + 1$. Find $f(-2)$.
Step 1. The input is given and the output is wanted, so this is a substitution — no equation to solve.
Step 2. $f(-2) = 5(-2) + 1 = -10 + 1 \Rightarrow$ $\mathbf{-9}$

Try it: $f(x) = 5x + 1$. Find $f(4)$.

check: **21**

2. Finding the input for an output

When the *output* is the number you are given, set the rule equal to it and solve for the input. Evaluating and solving are the same rule run in opposite directions, so each one checks the other.

Example: $f(x) = 5x + 1$. For what x is $f(x) = 26$?
Step 1. The 26 is an output, so it belongs on the right of an equation rather than inside the rule: $5x + 1 = 26$.
Step 2. Subtract 1: $5x = 25$. Divide by 5: $x = 5$. $\Rightarrow$ $\boldsymbol{x = 5}$ (check: $5(5) + 1 = 26$ ✓)

Try it: $f(x) = 5x + 1$. For what x is $f(x) = -14$?

check: $\boldsymbol{x = -3}$

3. Reading a table or a graph

Graph: to find $f(a)$, go across to a on the input axis, then straight to the plotted point and read the output axis. To find the input for a known output, go the other way round.

Table: if the outputs rise by the same amount every time the input rises by 1, the function is linear. That constant rise is the multiplier, and one row then pins down the constant term.

Example: The table gives $q(x)$. Find a rule, then find $q(7)$.

x	1	2	3
$q(x)$	5	8	11

Step 1. The table stops at $x = 3$ but the question asks about 7, so a rule is needed — reading the table cannot answer it.
Step 2. The outputs climb by 3 each step, so $q(x) = 3x + b$; using $x = 1$ gives $5 = 3 + b$, so $b = 2$ and $q(x) = 3x + 2$.
$\Rightarrow$ $q(7) = 3(7) + 2 = \mathbf{23}$

Try it: $r(x)$ is 9, 13, 17 at $x = 1, 2, 3$. Find $r(6)$.

check: **29**

4. Domain and range in a story

A bare rule usually accepts every number. A *story* does not: time starts at 0, a tank stops when it is empty, tickets come in whole numbers, an order has a limit. Find where the story ends by setting the rule equal to the value that ends it, then solving.

Example: A tank holds $W(t) = 300 - 20t$ litres t minutes after the valve opens, and the rule stops describing it once the tank is empty. Where does the domain end?
Step 1. The algebra would run forever, so the story sets the limit: empty means an output of zero.
Step 2. $300 - 20t = 0$, so $20t = 300$ and $t = 15$.
$\Rightarrow$ $\boldsymbol{t = 15}$, so a sensible domain is $0 \leq t \leq 15$.

Try it: $B(t) = 250 - 25t$ litres, same story. Where does the domain end?

check: $\boldsymbol{t = 10}$

(!) Watch out: $f(3)$ never means $f \times 3$. And two inputs may share one output (the drone is at 40 feet twice), but one input may never have two outputs — that is what stops a picture from being a function.